

Failure mode measured on the VCS	VCS measured evidence proven in this benchmark		Neural-network analogue documented, not proven here	Status here → there
1 Gradient vanishes on discrete / index outputs	0.000 position-regime F single record	Vanilla saliency & IG = 0.000 on the position regime (provably zero at a strobe-timed argmax); whole gradient bucket 0.068.	Shattered / saturated gradients; non-differentiable argmax & index readouts.	proven here analogue there
2 Saliency is model-invariant (sanity-check fails)	0.000 sanity-check margin single record	Program-randomization: saliency cosine = 1.000 while 95% of RAM changed → sanity check FAILS.	Saliency unchanged under model- & label-randomization; behaves like an edge detector.	proven here analogue there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F 95% CI [0.02, 0.29]	Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116.	Spurious Granger edges; 'present ≠ used' tuning & attention read as explanation.	proven here analogue there
4 Single-unit ablation has an interaction blind-spot	0.414 connectome F 95% CI [0.08, 0.78]	Single-unit lesion is faithful (0.987) yet misses the wiring: connectome recovered at only 0.414.	Single-unit ablation misleading; computation is distributed across many units.	proven here analogue there
5 SAE / probe: present-not-used + polysemanticity	0.162 selectivity F 95% CI [0.08, 0.24]	SAEs match a known variable yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are not causal.	Control-task selectivity gap; superposition / polysemantic neurons.	proven here analogue there
6 Baseline-sensitivity of Integrated / Expected Gradients	0.034 EG F (all-regime) 95% CI [0.00, 0.10]	IG magnitude is baseline-sensitive (up to 0.999 across games); a content-bytebaseline kills IG; EG lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input.	proven here analogue there

■ VCS evidence: proven in this benchmark (vs the exact \$3.2 intervention oracle) 0.344 faithfulness gap; activation patching = 1.000 (oracle ceiling) where vanilla saliency = 0.000 (naive-gradient floor).
■ NN analogue: documented in the literature, not proven by this benchmark